



Original Article

Numerical investigation of spreading phenomena of molten corium using EISPH method

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ABSTRACT

Understanding the behavior of molten corium is crucial for effective heat management and ensuring the integrity of reactor containment during severe nuclear incidents. Corium spreading is a complex multi-physics phenomenon influenced by hydrodynamics, heat transfer, and phase change. The Smoothed Particle Hydrodynamics (SPH) method is utilized to study corium flow, leveraging its ability to handle complex fluid dynamics and its compatibility with parallel computing architectures. This research examines two specific corium spreading cases: the VULCANO VE-U7 experiment, which is characterized by a wide mushy zone, and the FARO L26S experiment, noted for its narrow mushy zone. The study compares spreading lengths and temperature profiles over time with experimental data, offering a detailed analysis of the observed spreading behaviors. Variations in physico-chemical properties result in distinct spreading patterns between the cases, highlighting the complexity of corium behavior under different scenarios. The findings demonstrate the applicability and capability of the SPH method in capturing various characteristics of corium spreading effectively.

1. Introduction

The operation of nuclear reactors involves the generation of decay heat within nuclear fuels even after reactor shutdown, whether manual or automatic. To manage this residual decay heat, nuclear facilities are equipped with robust heat removal systems that effectively transfer the heat from the reactor core to the ultimate heat sink. However, unexpected failures in these systems, exemplified by the Fukushima accident, can lead to prolonged ineffectiveness in decay heat removal. This results in the accumulation of thermal energy, which may cause nuclear fuel to melt down and potentially precipitate severe accidents [1]. To mitigate the risk of damage to concrete structures from such events, global nuclear reactors implement two principal strategies. The first strategy, In-Vessel Retention (IVR) augmented by External Reactor Vessel Cooling (IVR-ERVC), is designed to contain the molten corium within the reactor pressure vessel (RPV) while enabling external cooling. Although there is some controversy regarding the thermal safety margins of this strategy, it has been adopted in high-power reactors such as the APR1400 and APR + due to its efficacy in maintaining the integrity of the reactor vessel [2]. The second strategy, activated upon IVR failure, involves the transfer of corium to a flat or curved surface to facilitate controlled spreading and prolonged cooling, commonly known as the core catcher

method. This method is prevalent in reactors employing dry cavity designs, including the EPR, ESBWR, VVER, and EU-APR1400 models [2,3].

Regardless of the strategy adopted, a thorough understanding of the complex spreading behavior of corium, particularly its underlying mechanisms involving fluid dynamics, heat, and mass transfer, is essential for both approaches. Nevertheless, experiments on corium spreading, which typically utilize prototypical corium compositions including depleted uranium oxide, are both complex and costly. Initially, these experiments employed simulant materials such as CORINE [4], S3E [5], SPREAD [6], KATS [7], and ECOKATS [8], which, despite their practicality, fail to accurately replicate critical corium phenomena like the extensive liquidus-solidus temperature range and specific chemical properties [9]. In response to these limitations, more comprehensive studies using prototypical materials, such as those in the COMAS [10], FARO [11], and VULCANO [12] series, have been initiated to provide deeper insights into corium spreading behaviors.

To address the challenges posed by high temperatures and complex material compositions in corium spreading experiments, various numerical methods have been developed. These methods are cost-effective, allow for large-scale scenario modeling, and generate extensive data throughout the simulation process, thereby overcoming the limitations of physical experiments. Initially, numerical approaches to corium

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spreading relied on semi-analytical methods [5,49,50]. However, these methods could only provide limited information, such as spreading length, and were confined to simple cases like isothermal spreading. Consequently, the development of lumped parameter codes such as MELTSPREAD [13], THEMA [14], LAVA [15], CROCO [16], and CORFLOW [17] are developed and utilized to study corium spreading phenomena. These lumped parameter approaches have closely matched experimental results through empirical correlations and can provide comprehensive information. However, they face difficulties in delivering detailed and mechanistic insights, especially when using one-dimensional or two-dimensional approaches.

Recent advancements in computational technology have promoted the use of mesh-based or meshless CFD analysis for examining corium spreading behaviors. Commercial CFD such as ANSYS CFX and FLUENT have been employed to handle the multiphase dynamics of these processes, incorporating temperature-dependent viscosity models to simulate corium solidification [18,19]. The Finite Volume Method (FVM) is utilized alongside the Volume-of-Fluid (VOF) model to manage the interface between air and melt [20]. Similarly, the Finite Element Method (FEM) is adopted in conjunction with the Conformal Decomposition Finite-Element Method (CDFEM) to monitor the corium/air interface using a background mesh [21].

The particle methods such as Moving Particle Semi-implicit (MPS) method or SPH have gained traction due to those inherent strength on managing free surfaces without relying on complex numerical methods. Initial applications using the MPS method demonstrated significant discrepancies in results due to the disregarding of radiative heat transfer [22]. These discrepancies were later addressed through the integration of a radiative heat transfer model, achieving better correlation with specific experiments [23]. Recent developments include the introduction of an implicit viscosity solver for efficient and accelerated simulation on sudden increases in viscosity and immobilization of flows [24, 25]. Recent studies have taken a sophisticated approach to studying spreading phenomena. Some studies have focused on calculations of Mises stress to model the stop-and-go phenomenon in ECOKATS-V1 experiment [26]. Additionally, numerical investigations have been performed to simulate lubricant effects from the interaction between molten corium and concrete substrates which has been observed in VULCANO VE-U9 experiment by incorporating slip boundary conditions [27]. Different from previous studies conducted using the MPS method, the most recent study replicated a well-known experiment using a 2D ISPH method with a novel boundary treatment technique [28].

A variety of meshless and mesh-based methods have been employed to investigate corium spreading phenomena. However, most studies have primarily focused on examining superficial behaviors that have traditionally been analyzed through lumped parameter approaches. In this study, the Explicit Incompressible Smoothed Particle Hydrodynamics (EISPH) method is utilized not only to assess these apparent properties but also to perform a comprehensive analysis of both local and global spreading behaviors, capitalizing on the detailed capabilities of CFD analysis. To enhance the stability of the numerical model, the governing equations have been designed using a quasi-Lagrangian framework, and an implicit viscosity solver has been integrated to expedite computational processes. Additionally, the entire algorithm has been parallelized on GPUs to augment efficiency.

The structure of the remaining sections of this paper is as follows: **Section 2** provides an overview of the numerical method employed. **Section 3** introduces the benchmark problems examined in this study: VULCANO VE-U7 and FARO-L26S. **Section 4** presents and discusses the numerical results for both benchmarks in detail. Finally, conclusions are summarized and future research directions are suggested in **Section 5**.

2. Numerical method

In this section, we present a brief explanation on SPH discretization and governing equations for simulating corium spreading phenomena

are presented.

2.1. SPH approximation

For any scalar function $f(\mathbf{r})$, its value at a point $\mathbf{r}$ in space is approximated by convolving this function over a support domain Ω using a weighting function $W(\mathbf{r}-\mathbf{r}', h_s)$ with a smoothing length h_s as follows [29]:

$$f(\mathbf{r}) = \int_{\Omega} f(\mathbf{r}') W(\mathbf{r}-\mathbf{r}', h_s) d\mathbf{r}'. \quad (1)$$

In the discretized SPH form, where the fluid is represented by particles, the value of f at the location of the i -th particle, denoted $\mathbf{r}_i$, is given by Ref. [29]:

$$f_i = f(\mathbf{r}_i) = \sum_j f_j W_{ij}(|\mathbf{r}_{ij}|, h_{s,i}) V_j. \quad (2)$$

where $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$ represents the relative position of neighboring particles, $V_j = m_j/\rho_j$ denotes the volume associated with the j -th particle, and W_{ij} is the weighting function simplified for notation. For this study, we employ the Wendland C2 kernel function as the weighting function, renowned for its outstanding stability [30,31]:

$$W(q) = \begin{cases} C(1-q)^4(1+4q), & \text{if } 0 \leq q < 1 \\ 0, & \text{if } q \geq 1 \end{cases} \quad (3)$$

Here, $q = r/(\kappa h_s)$, $C = 21/(2\pi(\kappa h_s)^3)$ in 3D, and κ as a scale factor (set to 1.666), and the smoothing length $h_s = 1.5\Delta x$, where Δx is the initial particle spacing. The overall support radius of particle can be computed as κh .

The computation of the gradient, divergence, and Laplacian of scalar and vector fields in SPH is crucial for enforcing conservation laws and simulating physical behaviors. The SPH formulations for SPH gradient, and Laplacian operators for a scalar function f_i and the divergence operators for a vector function $\mathbf{F}_i$ are outlined as follows [32]

$$\nabla f_i = \sum_j [f_j \pm f_i] \nabla_i W_{ij} V_j, \quad (4)$$

$$\nabla \cdot \mathbf{F}_i = \sum_j [\mathbf{F}_j \pm \mathbf{F}_i] \cdot \nabla_i W_{ij} V_j, \quad (5)$$

$$\nabla^2 f_i = \sum_j 2[f_j - f_i] \frac{\mathbf{r}_{ij} \cdot \nabla_i W_{ij} V_j}{|\mathbf{r}_{ij}|^2}. \quad (6)$$

2.2. Governing equations

In the Lagrangian description of fluid dynamics, the governing equations for the motion of an incompressible, and Newtonian fluid encompass conservation of mass, momentum, and energy which are written as follows:

$$\nabla \cdot \mathbf{u} = 0, \quad (7)$$

$$\frac{D\mathbf{u}}{Dt} = -\frac{\nabla P}{\rho} + \nu \nabla^2 \mathbf{u} + \mathbf{g} + \sigma \kappa \mathbf{n} \delta_s, \quad (8)$$

$$\frac{Dh}{Dt} = \frac{k_i}{\rho_i} \nabla^2 T + \frac{Q_{rad}'''}{\rho_i}. \quad (9)$$

In the given expressions, $\mathbf{u}$ represents the velocity, while $\frac{D}{Dt}$ denotes the material derivative. The term P signifies the fluid pressure, and ρ , the density of the fluid, is considered constant owing to the incompressibility assumption. Furthermore, ν is the kinematic viscosity, and $\mathbf{g}$ represents the gravitational force exerted on the fluid. σ is surface tension, κ is the curvature, $\mathbf{n}$ is the normal vector and δ_s is the interfacial function. Additionally, h corresponds to specific enthalpy, T indicates the

temperature, and Q_{rad}^m refers to the rate of heat loss via radiation per unit volume.

2.3. EISPH

In the present study, the EISPH has been chosen for solving governing Equations (7)–(9) due to its established stability and robustness. The discretized governing equations of EISPH framework in fluid dynamics analysis for multi-phase flows include the Pressure Poisson Equation (PPE) and the momentum equation [33]:

$$\sum_{j=1}^N \left(p_i^{n+1} - p_j^n \right) \frac{2\mathbf{r}_{ij} \bullet \nabla W(\mathbf{r}_{ij}, h) V_j}{|\mathbf{r}_{ij}|^2 + \eta^2} = \frac{\sum_j \left(\frac{2\rho_i \rho_j}{\rho_i + \rho_j} \right) \mathbf{u}_{ji}^* \bullet \nabla W(\mathbf{r}_{ij}, h) V_j}{\Delta t}, \quad (10)$$

$$\frac{D\mathbf{u}_i}{Dt} = \mathbf{a}_{pres} + \mathbf{a}_{vis} + \mathbf{a}_{remain}, \quad (11)$$

where $\mathbf{u}_{ji}$ denotes the differences $\mathbf{u}_j - \mathbf{u}_i$ and η is a small constant set to be $0.1h$ to avoid singularity problem. $\mathbf{a}_{pres}$, $\mathbf{a}_{vis}$, and $\mathbf{a}_{remain}$ represent the acceleration from pressure force, viscous force, and the remaining force excluding the pressure force and viscous force from the total force, respectively. The remaining forces are those excluding the contributions from both pressure and viscous forces within the total force acting on the fluid, which composed of gravitational force and surface tension force in this study.

$$\mathbf{a}_{remain} = \mathbf{g} - \sigma \kappa_i \mathbf{n}_i \delta_{s,i}. \quad (12)$$

$$\mathbf{a}_{vis} = \sum_{j=1}^N \left(\frac{2\nu_i \nu_j}{\nu_i + \nu_j} \right) (\mathbf{u}_i - \mathbf{u}_j) \frac{2\mathbf{r}_{ij} \bullet \nabla_i W_{ij} V_j}{|\mathbf{r}_{ij}|^2 + \eta^2}. \quad (13)$$

$$\mathbf{a}_{pres} = -\frac{1}{\rho_i} \sum_j 2 \left(\frac{P_i \rho_i + P_j \rho_j}{\rho_i + \rho_j} \right) \nabla_i W_{ij} V_j. \quad (14)$$

The computations of the normal vector, and curvature are detailed in previous study [34]. For further information, please refer to their work. The interfacial function is defined as:

$$\delta_i = 2 \left| \sum_{j \in \Omega} V_j \nabla_i W_{ij} \right|. \quad (15)$$

In this study, the energy conservation equation is also addressed, and the discretized form of the energy equation is given as follows:

$$\frac{Dh_i}{Dt} = \frac{1}{\rho_i} \sum_j \left(\frac{2k_i k_j}{k_i + k_j} \right) h_{ji} \frac{2\mathbf{r}_{ij} \bullet \nabla_i W_{ij} V_j}{|\mathbf{r}_{ij}|^2 + \eta^2} - \frac{1}{\rho_i} [\sigma \varepsilon_i (T_i^4 - T_\infty^4)] \delta_{s,i}. \quad (16)$$

In this expression, σ is Stefan-Boltzmann constant, ε is emissivity, T_∞ represents ambient temperature. The overall procedure of solving discrete governing equations utilizing the EISPH methodology can be succinctly summarized into following steps [35]:

- i. Calculation of acceleration excluding the pressure force, expressed as $\mathbf{a}_{vis} + \mathbf{a}_{remain}$.
- ii. Computation of energy equation [Eq. (16)]
- iii. Computation of the first intermediate velocity $\mathbf{u}_i^* = \mathbf{u}_i^n + (\mathbf{a}_{vis} + \mathbf{a}_{remain}) \Delta t$.
- iv. Computation of the Pressure Poisson Equation (PPE) [Eq. (10)].
- v. Calculation of the pressure force $\mathbf{a}_{pres}$.
- vi. Computation of the corrected velocity $\mathbf{u}_i^{n+1} = \mathbf{u}_i^* + \mathbf{a}_{pres} \Delta t$.
- vii. Calculation of the corrected position $\mathbf{r}_i^{n+1}$.

For the solid particles, they are modeled as fixed particles with densities identical to those of the fluid particles. The viscosity is set to match that of fully solidified corium. To enforce the Neumann pressure boundary condition and the no-slip velocity boundary condition, a

generalized boundary condition (GBC) —one of the most common boundary conditions in the SPH community—has been adopted as the default [51]. In GBC, the pressure and velocity in solid particles are computed based on pressure and velocity field of fluid particles as follows:

$$P_{wall} = \frac{\sum_{j \in \Omega_f} P_j W_{ij} + (\mathbf{g} - \mathbf{a}_{wall}) \bullet \sum_{j \in \Omega_f} \rho_j \mathbf{r}_{ij} W_{ij}}{\sum_{j \in \Omega_f} W_{ij}}, \quad (17)$$

$$\mathbf{u}_{wall} = 2\mathbf{U}_{wall} - \frac{\sum_{j \in \Omega_f} \mathbf{u}_j W_{ij}}{\sum_{j \in \Omega_f} W_{ij}}. \quad (18)$$

Here, Ω_f is the fluid domain. In our study, both the desired wall velocity ($\mathbf{U}_{wall}$) and wall acceleration ($\mathbf{a}_{wall}$) are set to zero.

Additionally, another velocity boundary condition considered in our study is known as the Continuous Boundary Force (CBF) method, which we have slightly modified [48]. In this method, the no-slip boundary condition is enforced in a more comprehensive form, like the Navier slip boundary condition. This is achieved by transforming the boundary condition into a volumetric force term applied to the fluid particles as following term, ensuring that the motion of the fluid near the boundary adheres closely to the specified no-slip or partial slip conditions dictated by the simulation requirements.

$$\mathbf{a}_{CBF} = -\beta \delta_{s,i} \mathbf{u}_i \quad (19)$$

The interfacial function for CBF is defined as:

$$\delta_{s,i} = 2 \left| \sum_{j \in \Omega_s} V_j \nabla_i W_{ij} \right|. \quad (20)$$

Here, Ω_s is the solid domain, $\beta = \mu/b$ represents the friction coefficient, which characterizes the interaction at the fluid-solid interface, and b is the slip length. Therefore, when β is 0 or b goes to infinite, it indicates a free slip condition, implying no friction at the surface, whereas a β tending towards infinity or b goes to zero corresponds to a no-slip condition. Due to computationally manageability, β cannot be infinity but must be chosen as a suitable large value. In this study, β ranges from 10^3 to 10^5 to effectively enforce the no-sip boundary conditions and further discussed in section 4.

2.4. Quasi-Lagrangian formulation of EISPH

In this study, the governing equations are solved using a quasi-Lagrangian formulation, benefiting from insights gained in previous research that highlighted improvements in stability and accuracy [36, 37]. For this reformulation, the governing equations are re-expressed utilizing the Arbitrary Lagrangian-Eulerian (ALE) time derivative, defined as:

$$\frac{df}{dt} := \frac{\partial f}{\partial t} + (\mathbf{u} + \delta \mathbf{u}) \bullet \nabla f = \frac{Df}{Dt} + \delta \mathbf{u} \bullet \nabla f. \quad (21)$$

In this context, $\partial t / \partial$ represents the Lagrangian derivative, while $\delta \mathbf{u}$ denotes the particle shifting velocity. If $\delta \mathbf{u}$ equals zero, the system operates entirely under Lagrangian conditions. The particle shifting velocity is calculated using a modified concentration gradient, according to Fick's law, to ensure particle uniformity and mitigate tensile instability [38].

$$\nabla_i C_i = \sum_j \left[1 + 0.2 \left(\frac{W_{ij}}{W_{ij}(\Delta x)} \right)^4 \right] \frac{m_j}{\rho_j} \nabla_i W_{ij} \quad (22)$$

$$\delta \mathbf{u}_i = -D_i' \nabla_i C_i \quad (23)$$

In this paper, the velocity displacement of interior particle is derived as [[39,40]]:

$$\delta \mathbf{u}_i = \delta \mathbf{u}_{i,interior} := -\kappa h_i |\mathbf{u}_i|_{max} \nabla_i C_i. \quad (24)$$

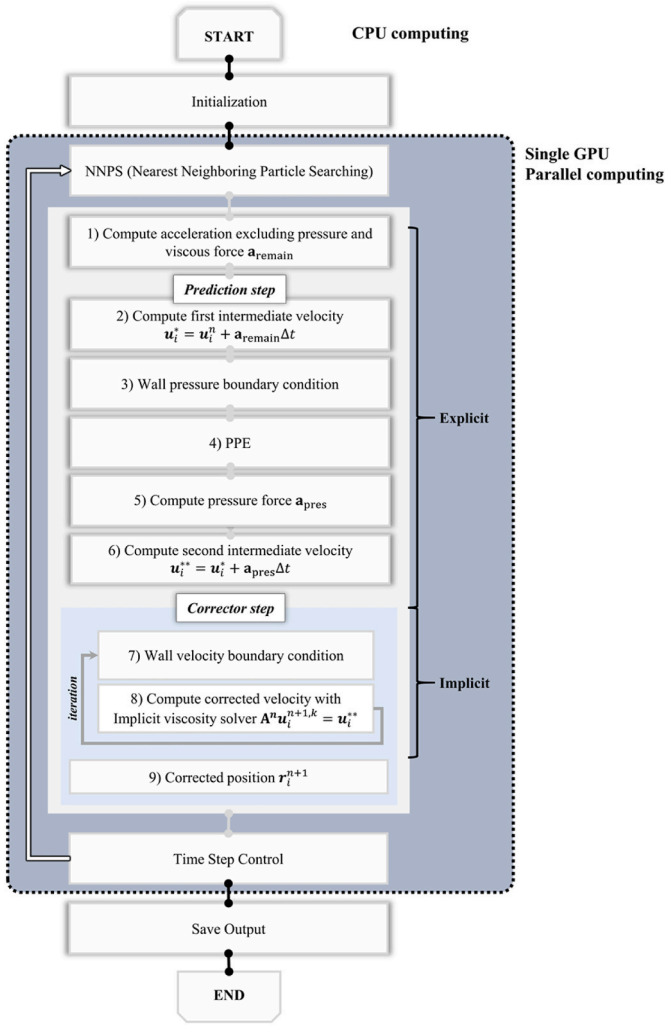


Fig. 1. Overall coding structure.

In order to maintain the kinematic boundary conditions at interfaces and prevent distortion of the flow, particles near free surface regions are shifted in accordance with their physical velocity, resulting in a shifting velocity ($\delta \mathbf{u}_i$) of zero. Free surface particles are identified using a specific criterion that involves the minimum eigenvalue (λ_i) of the inverse of the renormalization matrix (L_i^{-1}) [41]. Particles with a λ_i value less than 0.6 are classified as free surface particles. Particles that are neither strictly free surface nor completely interior, termed as interface particles, are defined as those having λ_i values between 0.6 and 0.8. To manage these interface particles effectively, an optimized particle shifting strategy is employed [42]. This strategy involves eliminating components in the normal direction and permitting movement solely in the tangential direction to ensure accurate representation and dynamics at the interface.

$$\delta \mathbf{u}_i = \delta \mathbf{u}_{i, \text{interface}} : = -\kappa h_i |\mathbf{u}_i|_{\max} (\mathbf{I} - \mathbf{n}_i \otimes \mathbf{n}_i) \cdot \nabla_i C_i. \quad (25)$$

To prevent the particle shifting velocity $\delta \mathbf{u}_i$, from exceeding half of the physical velocity $|\mathbf{u}_i|$, following constraint is applied:

$$\delta \mathbf{u}_i = \begin{cases} \delta \mathbf{u}_i & \text{if } |\delta \mathbf{u}_i| < 0.5 |\mathbf{u}_i|, \\ -0.5 |\mathbf{u}_i| \frac{\delta \mathbf{u}_i}{|\delta \mathbf{u}_i|} & \text{otherwise,} \end{cases} \quad (26)$$

Within the ALE framework, and considering the shifting velocity, it is necessary to account for the deviation from purely Lagrangian motion by including additional convection terms ($\delta \mathbf{u}_i \cdot \nabla_i \mathbf{u}_i$ and $\delta \mathbf{u}_i \cdot \nabla_i T_i$ in Equations 11 and 16. This study also incorporates upwind interpolation

for the convection terms, to enhance numerical stability and accuracy as supported [[39,40]]. In upwind interpolation, the following anisotropic weight function is used to evaluate the convective terms [40]:

$$W_{ij}^u = \begin{cases} W_{ij} & \theta_{ij} \geq \pi/2 \\ \epsilon W_{ij} & \theta_{ij} < \pi/2 \end{cases}. \quad (27)$$

where ϵ is small positive value and set as 0.01 in this study, θ_{ij} is the angle between $\mathbf{r}_{ij}$ and the direction of shifting velocity:

$$\theta_{ij} = \arccos \left(\frac{\mathbf{r}_{ij}}{\|\mathbf{r}_{ij}\|} \cdot \frac{\delta \mathbf{u}_i}{\|\delta \mathbf{u}_i\|} \right). \quad (28)$$

The final governing equations are discretized as follows:

$$\frac{d\mathbf{u}_i}{dt} = \mathbf{a}_{\text{pres}} + \mathbf{a}_{\text{vis}} + \mathbf{a}_{\text{remain}}, \quad (29)$$

$$\frac{dh_i}{dt} = \frac{1}{\rho_i} \sum_j \left(\frac{2k_i k_j}{k_i + k_j} \right) h_{ji} \frac{2\mathbf{r}_{ij} \cdot \nabla_i W_{ij} V_j}{|\mathbf{r}_{ij}|^2 + \eta^2} + \delta \mathbf{u}_i \cdot \sum_j h_{ji} \nabla_i W_{ij} \frac{m_j}{\rho_j} - \frac{1}{\rho_i} [\sigma \epsilon_i (T_i^4 - T_\infty^4)] \delta_{S,i}, \quad (30)$$

where $\mathbf{a}_{\text{remain}}$ is rewritten as:

$$\mathbf{a}_{\text{remain}} = \mathbf{g} - \sigma \kappa_i \mathbf{n}_i \delta_{S,i} + \delta \mathbf{u}_i \cdot \sum_j \mathbf{u}_{ji} \otimes \nabla_i W_{ij} \frac{m_j}{\rho_j}. \quad (31)$$

2.5. EISPH with implicit time integration of viscous force

In the simulation of molten corium solidification, a significant increase in viscosity restricts the allowable time step. To overcome the limitation posed by the viscous force, the component of viscous force is handled implicitly. This modified approach involves a two-stage computation of intermediate velocity. In this modified approach, the computation of the intermediate velocity occurs in two stages. Initially, the intermediate velocity is calculated using the acceleration attributable to forces excluding the pressure and viscous forces. Subsequently, the pressure field is determined based on this first intermediate velocity, and the pressure force is calculated. Thereafter, utilizing the acceleration from the pressure force, the second intermediate velocity is updated. The viscous force is then implicitly resolved using this second intermediate velocity. The modified algorithm for the EISPH method with an implicit viscosity solver is summarized as follows.

- i. Calculation of acceleration excluding the pressure force and viscous force $\mathbf{a}_{\text{remain}}$.
- ii. Computation of energy equation [Eq. 16]
- iii. Computation of the first intermediate velocity $\mathbf{u}_i^* = \mathbf{u}_i^n + (\mathbf{a}_{\text{remain}}) \Delta t$.
- iv. Computation of the Pressure Poisson Equation (PPE).
- v. Calculation of the pressure force $\mathbf{a}_{\text{pres}}$.
- vi. Compute of the second intermediate velocity $\mathbf{u}_i^{**} = \mathbf{u}_i^* + \mathbf{a}_{\text{pres}} \Delta t$.
- vii. Computation of the corrected velocity implicitly $\mathbf{u}_i^{n+1} = \mathbf{u}_i^{**} + \mathbf{a}_{\text{vis}} \Delta t$.
- viii. Calculation of the corrected position $\mathbf{r}_i^{n+1}$.

The relationship between the second intermediate velocity and corrected velocity is written as:

$$\mathbf{u}_i^{n+1} = \mathbf{u}_i^{**} + \mathbf{a}_{\text{vis}} \Delta t = \mathbf{u}_i^{**} + \frac{1}{\rho_i} \sum_{j=1}^n \mu_{ij}^n (\mathbf{u}_i^{**} - \mathbf{u}_j^{**}) \frac{2\mathbf{r}_{ij} \cdot \nabla_i W_{ij} V_j}{|\mathbf{r}_{ij}|^2 + \eta^2}. \quad (32)$$

In order to solve this Eq. (32) implicitly, the conjugate gradient method with Jacobi preconditioner is utilized in matrix-free formulation. Due to the implementation of implicit viscosity solver, the time step is free from constraint induced by viscous dissipation and limited only by the maximum speed and thermal diffusion:

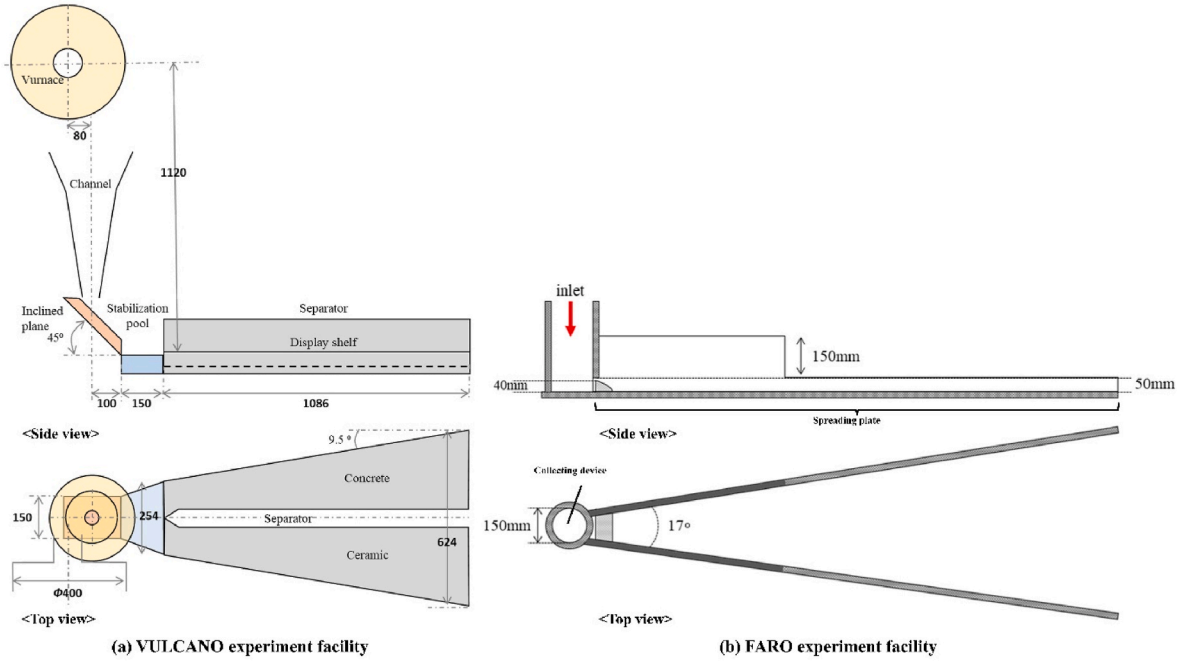


Fig. 2. Schematic view of VULCANO (a) and FARO (b) Experiment Facilities.

$$\Delta t \leq \left(CFL \min \left(\frac{h_s}{\|u_i\|}, 0.125 \frac{h_s^2}{\alpha} \right), \right) \quad (33)$$

In this study, the *CFL* is set to be 0.1 and α represents the thermal diffusion.

2.6. Coding algorithm

For this study, the SOPHIA code developed by Seoul National University is utilized as the foundational structural framework [43,44,52]. The overall algorithm in this study is structured around CPU-GPU heterogeneous computing as shown in Fig. 1. The initialization step, which involves reading inputs, material properties, setting parameters, etc., and the output saving step, are executed on the CPU. Meanwhile, the solvers for the overall governing equations are parallelized on a single GPU. In the solver, the integration of viscous forces is performed implicitly with respect to time, while the remaining parts are computed explicitly. For computation, a single NVIDIA A100 80 GB PCIe GPU and an Intel Xeon Gold 6342 CPU @ 2.80 GHz are utilized in this study.

3. Numerical setup for corium spreading experiments

In this section, we present an explanation of selected benchmark corium spreading experiments and the numerical setup for numerical investigation. In this study, the VULCANO VE-U7 [12,45] and FARO-L26S [11] experiments were selected as benchmarks due to the detailed information on experimental condition and results are provided for public and their frequent use as benchmark examples in other research papers. The schematic representation of both experiments facilities are demonstrated in Fig. 2.

3.1. VULCANO VE-U7: experimental condition

The VULCANO VE-U7 experiment is one of the experiments conducted using the VULCANO facility at CEA in France and focused on the dynamics of corium spreading, particularly examining the influence of concrete on this phenomenon [45]. In this experimental setup, a corium composition similar to that hypothesized for the EPR containment vessel

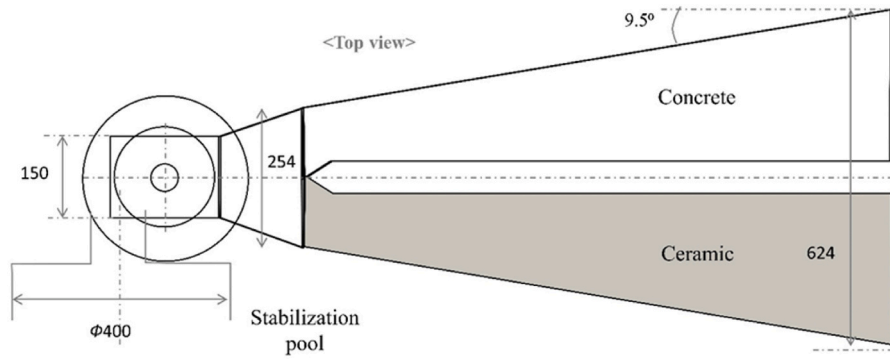
was utilized, consisting of UO₂ (56 wt%), ZrO₂ (32 wt%), and other oxides [46]. The solidus-liquidus temperature range of the melt mixture, calculated using the GEMINI2 software, exceeds 1000 K, indicating a very wide range. This mixture was introduced into a test section that featured two spreading channels, each inclined at an angle of 9.5° and lined with different substrates: one with ceramic, made of dense electro-fused zirconia bricks, and the other with concrete as in Fig. 2a in the experiment, corium is poured into a furnace and then flattens in a stabilization pool to prevent dripping, subsequently bifurcating into both ceramic and concrete channels [45].

Approximately 40.8 kg of corium was discharged from the furnace at a rate of 4.3 kg/s into the test section, which was initially maintained at a temperature of 2450 ± 80 °C. The substrate temperatures were recorded at approximately 299 ± 2 °C. The time required for the corium to achieve immobilization varied between the channels, taking about 7.7 s in the concrete channel and 12 s in the ceramic channel. In terms of spread, 12.6 kg of corium covered a distance of approximately 36 cm on the concrete substrate, whereas 14 kg traveled about 46 cm on the ceramic substrate. The detailed information of these experimental conditions and results render them suitable for use as a benchmark example [45].

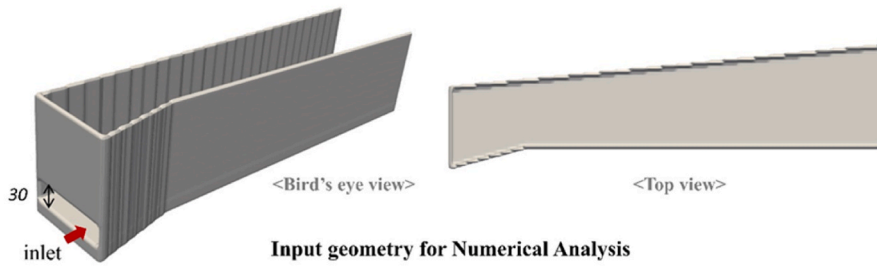
From the experimental results, the rates of ablation and decomposition of concrete were observed to be minor, occurring at a scale of a few millimeters-substantially smaller when compared to the temporal scale of corium spreading. Notably, the phase change of concrete is not accounted for in this analysis, as substantiated by prior researches [25, 45]. The validity of this assumption is further reinforced by the MELT-SPREAD simulation outcomes [13].

3.2. VULCANO VE-U7: numerical setup

As discussed in the previous subsection, the VULCANO VE-U7 experiment features both ceramic and concrete channels. Given that ceramic is inert and concrete reacts with molten corium, many numerical studies on corium spreading have focused solely on the ceramic channel. Recent work by Duan [25] has included analyses of both channels and conducted sensitivity studies on thermo-physical properties that may be affected by the interaction between molten corium and concrete (MCCI), such as effective conductivity, emissivity, and

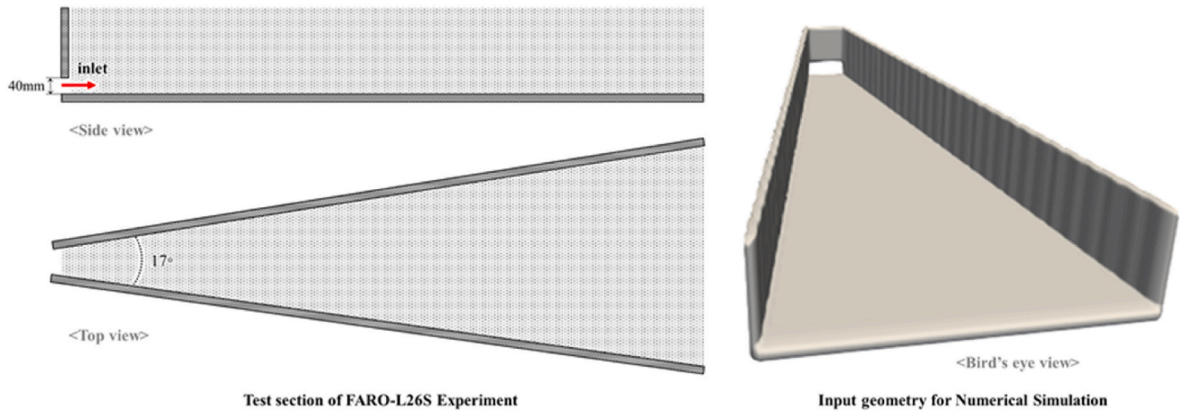


Test section of VE-U7 Experiment



Input geometry for Numerical Analysis

Fig. 3. Geometric information of the VULCANO VE-U7 experiment for numerical simulation [12].



Test section of FARO-L26S Experiment

Input geometry for Numerical Simulation

Fig. 4. Geometric information of the FARO-L26S experiment for numerical simulation [11].

Table 1
Material properties utilized in VE-U7 analysis [12].

	Corium	Ceramic substrate
Initial temperature [K]	2450	300
Solidus temperature [K]	1273	–
Liquidus temperature [K]	2623	–
Density [kg/m ³]	7140	5300
Thermal conductivity [W/m/K]	3.0	4.7
Specific heat [J/kg/K]	800	575
Emissivity	0.8	–
Fusion enthalpy [kJ/kg]	420	–
Viscosity of liquid [Pa•s]	0.0206	–

Table 2
Material properties utilized in FARO-L26S analysis [11].

	Corium	Steel substrate
Initial temperature [K]	2950	300
Solidus temperature [K]	2860	–
Liquidus temperature [K]	2910	–
Density [kg/m ³]	8000	8000
Thermal conductivity [W/m/K]	2.88	30.0
Specific heat [J/kg/K]	565	450
Emissivity	0.8	–
Fusion enthalpy [kJ/kg]	362	–
Viscosity of liquid (Pa•s)	0.0045	–
Surface tension (N/m)	0.45	–

viscosity. In this study, VE-U7 was analyzed, and only corium spreading on the ceramic substrate was modeled and interpreted, excluding the influence of MCCI as in Fig. 3. Continuous mass flow is supplied at the inlet, as detailed in the previous literature by Duan [24]. To manage the computational resources effectively, the memories of newly supplied corium particles are allocated from a predefined large buffer memory.

Specifically, 14 kg of corium is supplied at a mass flux rate of 1.75 kg/s, a figure derived from post-analysis of the experiment. The material properties utilized in the simulation, which include corium, ceramic, and concrete, are meticulously detailed in Table 1 (see Fig. 4).

The viscosity model employed in this study is the Ramacciotti model

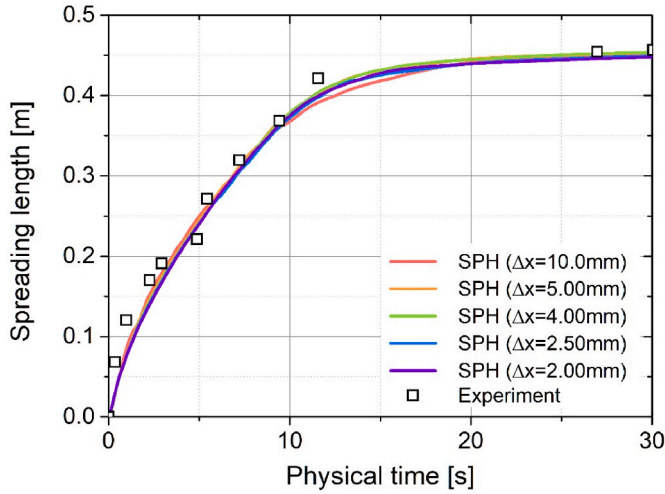


Fig. 5. Vulcano VE-U7: Time evolution of the position of the spreading front of corium with different particle resolutions.

[47], defined by the equation:

$$\mu(T) = \mu_{liq} e^{2.5C\gamma(T)}. \quad (34)$$

In this model, μ_{liq} represents the viscosity in the liquid phase, and $\gamma(T)$ denotes the solid fraction of the melt. The temperature-dependent solid fraction data were sourced data derived using the thermophysical codes GEMINI2 and TDBCRO01 [45]. The coefficient C in the Ramacciotti model typically ranges from 4 to 8 with experimental

alignments suggesting optimal results for values between 6 and 6.3; consequently, a value of 6 was adopted for this study [43]. The initial viscosity at the initial temperature ($T_{in} = 2450$ K) was set at 37 Pa-s, based on preliminary approximations [45].

However, the original viscosity model results in a continuous corium flow as the viscosity increases, without ceasing. It is known that corium motion nearly ceases when the solid fraction exceeds around 0.7 to 0.9 from the observation from VE-U7 experiment [12,45]. To accurately model the solidification of corium melt, a modification to the Ramacciotti model has been implemented [24,25]. The modified model is defined by the following expressions.

- For $\gamma(T) < \gamma_{imm}$, the viscosity is given by:

$$\mu(T) = \mu_{liq} e^{2.5C\gamma(T)}, \quad (35)$$

- For $\gamma(T) \geq \gamma_{imm}$, the viscosity is increased by a multiplication factor:

$$\mu(T) = M_{imm} \mu_{liq} e^{2.5C\gamma(T)}. \quad (36)$$

In numerical simulation of VE-U7, the immobilization solid fraction and multiplication factor is set as 0.8 and 10, respectively based on sensitivity studies from previous literature and a compromise for computational efficiency [24].

3.3. FARO-L26S: experimental condition

The FARO-L26S experiment, conducted at the FARO facility, investigates the spreading behavior of prototype corium with a UO₂/ZrO₂ weight percentage of 80/20. In this experiment, the corium mixture is poured into a furnace and then flows through an opening measuring 150

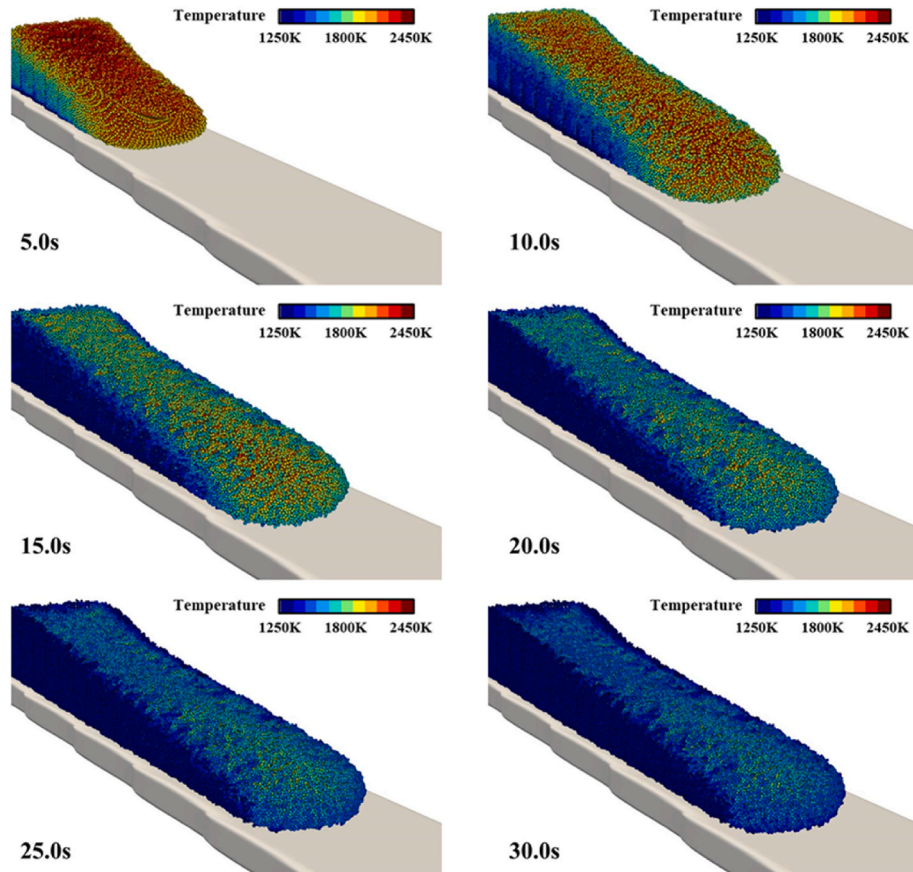


Fig. 6. VULCANO VE-U7: Time Evolution of Corium Progression (The colors of the particles represent their temperatures). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

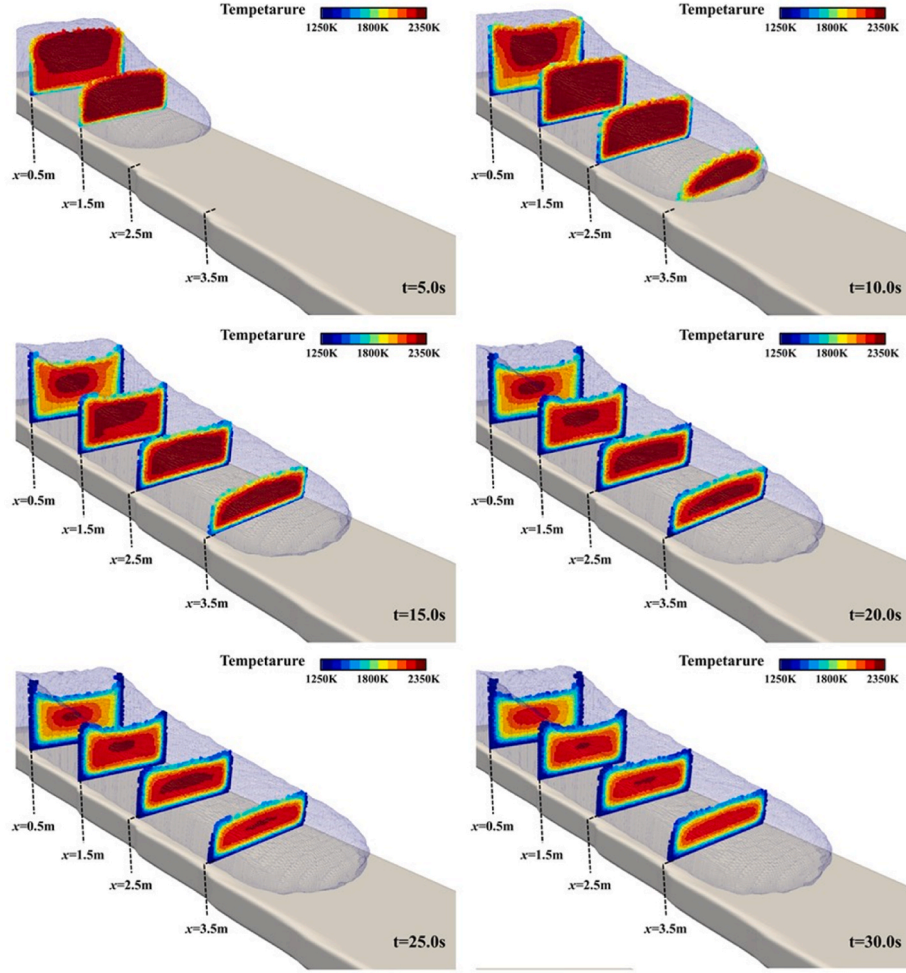


Fig. 7. VULCANO VE-U7: Time Evolution of Temperature Contours at $x = 0.5\text{m}$, 1.5m , 2.5m , and 3.5m Positions at Various Time Instants.

mm in width and 50 mm in height onto a spreading plate. During the spreading process, the mixture initially encounters a small structure with a height of 40 mm. Approximately 160 kg of high-temperature oxide melt at 2950 K is poured on a dry, non-coated stainless-steel plate forming a sector of 17° for 9.7 s [11].

3.4. FARO-L26S: numerical setup

In the FARO experiment, corium is poured into a furnace and then onto a spreading plate. At the entrance of the spreading plate, a 40 mm height pin obstructs the flow. However, due to the unavailability of detailed information regarding the amount of corium poured and the precise geometry of the pin, this study adopts a simplified geometry for modeling purposes [21]. In this study, the corium is assumed to be poured at a constant mass flux of 16.5 kg/s.

The thermophysical properties of the steel and prototype corium utilized in this study are detailed in Table 2. The solidus-liquidus temperature range of the melt mixture, calculated using the GEMINI2 software, is 50 K, which is very narrow and significantly differs from the conditions in the VULCANO experiments. The viscosity of the corium is modeled using the original Ramacciotti model without abrupt immobilization, and the solid fraction is given as linearly dependent on temperature [11].

4. Results and discussion

4.1. Numerical investigation on VULCANO VE-U7 experiment

To determine the appropriate particle resolution for simulations, experiments were conducted using a single-channel configuration with various particle resolutions. Initial particle spacings of 10 mm, 5 mm, 4 mm, 2.5 mm, and 2.0 mm were examined to monitor the progression of the ceramic spreading length over time. As shown in Fig. 5, increasing the spatial resolution led to satisfactory convergence in the trend of spreading length. Initially, cooling of the corium primarily occurs from the surface inward, predominantly through conduction and radiation, leading to a comparatively steep slope in the early stages as the bulk area remains less cooled (Phase 1). This stage is like isothermal spreading within a gravity-viscosity regime. As the cooling progresses, a significant increase in viscosity occurs in bulk region, ultimately ceasing the spreading behavior (Phase 2).

As the corium flows into the channel, cooling progresses from the surface to the interior due to conduction at the walls and radiation from the free surface, leading to a decrease in temperature. This decrease in temperature results in a substantial increase in viscosity, leading to the cessation of movement, as illustrated in the time sequence in Fig. 6. To enhance understanding of the spreading mechanism, Figs. 7–9 display the temperature, velocity, and solid fraction across different cross-sections over time, employing a particle spacing of 2.5 mm. Due to the sharp temperature decrease near the wall, caused by conduction and the significant temperature difference between the substrate and corium,

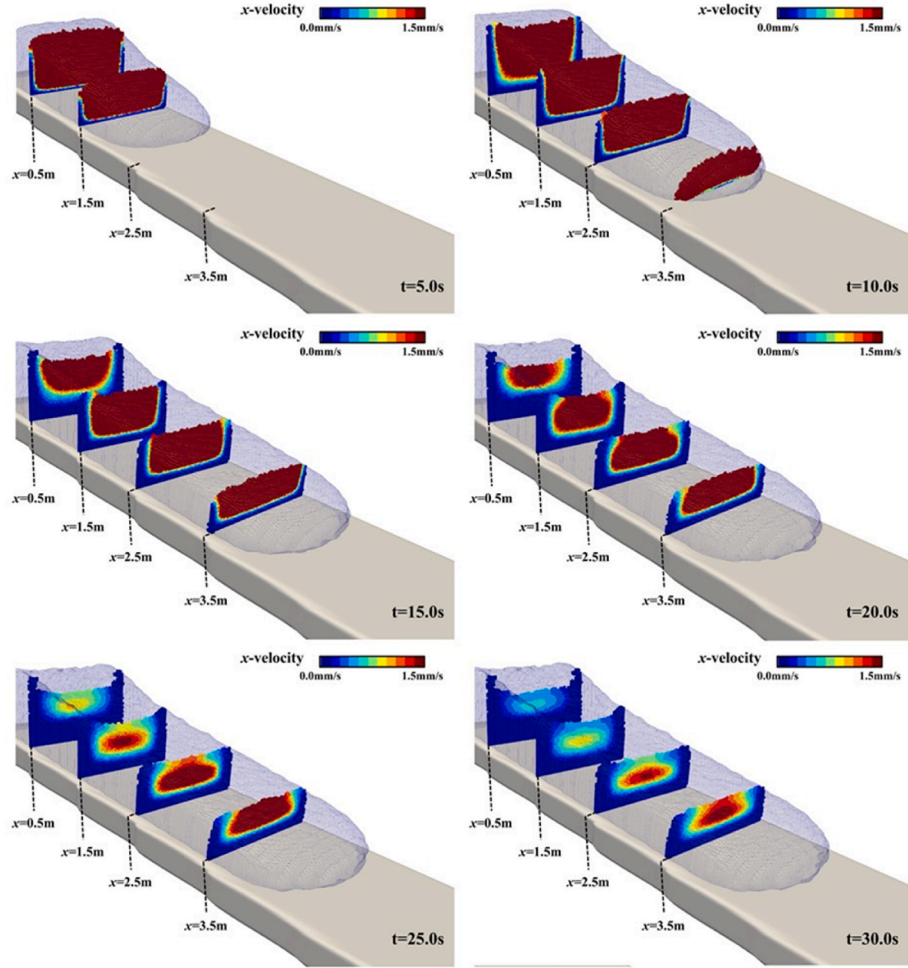


Fig. 8. VULCANO VE-U7: Time Evolution of Velocity Contours at $x = 0.5\text{m}$, 1.5m , 2.5m , and 3.5m Positions at Various Time Instants.

the phase change becomes more pronounced, leading to an increase in the solid fraction and a sharp rise in viscosity. As a result of the increased viscous dissipation, the flow may rapidly lose kinetic energy, which predominantly contributes to the halting of flow behavior near the wall boundary. However, the flow in the upward region maintains a relatively higher temperature, allowing the flow to sustain its continuity. Therefore, the cessation of global flow progression would be more influenced by radiative heat transfer rather than conductive heat transfer. This process eventually results in the development of a horizontally wrinkled shape, as observed in experimental data [45].

The evolution of temperatures are also monitored at $z = -7\text{mm}$ and $z = -12\text{mm}$ within the substrate, positioned 120 mm and 220 mm ahead of the corium injection site, as spatial resolution varied. The $z = -7\text{mm}$ location was specifically compared against experimental data. As illustrated in Fig. 10, while the results demonstrate good convergence except the coarsest case, they significantly deviate from experimental findings and results from LAVA code. This discrepancy has been noted in previous studies using both MPS methods and lumped parameter codes [24, 28, 45]. These studies attributed the variance to gap cooling and conducted further sensitivity analyses. However, in this study, the effect of gap cooling is not considered.

The calculations for amount of energy loss via different heat transfer mechanisms were performed, focusing on the time-dependent enthalpy flux per unit time due to conduction and radiation. As shown in Fig. 11, the curves representing enthalpy flux converge with increasing spatial resolution, except for the coarsest case, aligning with the trends observed in temperature. The flux rises steadily for the first 8 s following

corium injection and then gradually declines, attributed to cessation of injection and cooling of the material. This analysis indicates that conduction heat transfer surpasses that of radiation, likely because of a slightly larger area available for conduction, as illustrated by the time evolution contours in Figs. 7–9.

4.2. Numerical investigation on FARO-L26S experiment

In the FARO-L26S experiment, unlike the VULCANO VE-U7, the corium has a very narrow range between its solidus and liquidus temperatures. This narrow mushy zone leads to a rapid increase in viscosity as the temperature drops due to conduction and radiation, significantly faster than in the VULCANO VE-U7 scenario. Consequently, while the flow behavior of corium in the VULCANO VE-U7 experiment is relatively smooth, the motion in the FARO-L26S setting tends to be more erratic and complex. Therefore, as the corium progresses in the FARO experiment, due to rapid phase changes, some particles become isolated almost instantly.

To assign velocity boundary conditions on fluid particles, the CBF method is considered. With the GBC, no-slip is enforced by calculating and assigning the appropriate velocity to solid particles based on the velocity field of fluid particles near the solid boundary. This process can present challenges when isolated particles occur, as it becomes difficult to enforce no-slip accurately. Due to these challenges, our study has adopted the CBF method, which directly applies forces to fluid particles to strictly enforce no-slip conditions, ensuring more accurate and consistent adherence to the no-slip condition. This approach provides a

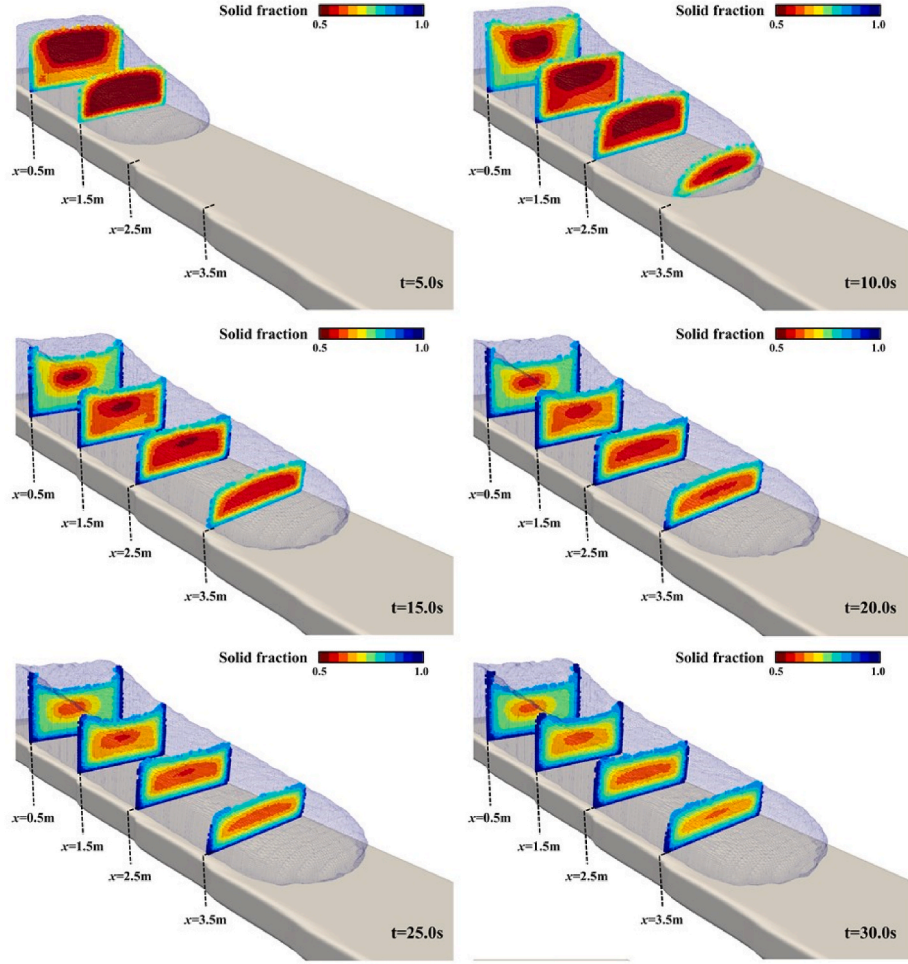


Fig. 9. VULCANO VE-U7: Time Evolution of Solid Fraction Contours at $x = 0.5\text{m}$, 1.5m , 2.5m , and 3.5m Positions at Various Time Instants.

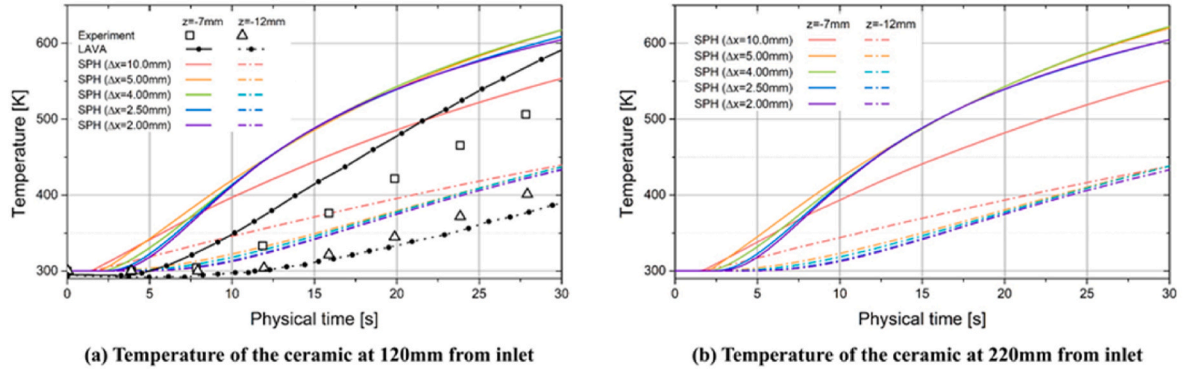


Fig. 10. Vulcano VE-U7: Time evolution of bottom wall temperature at different distances from the inlet with different particle resolutions (a) 120 mm (b) 220 mm.

robust solution to the limitations encountered with the GBC in handling dynamic and complex flow conditions in the FARO experiment.

In Fig. 12, the time evolution of the corium spreading front is depicted, showcasing simulations with different predefined slip lengths, denoted as b , and compared alongside experimental data. The particle spatial resolution in the simulation is set at 3 mm. When the slip length b is large, the corium quickly slips over the substrate, flowing rapidly without sufficient time for cooling via conduction heat transfer. As the slip length b decreases, slippage is increasingly restricted, leading to slower progression and more active cooling due to conduction, resulting

in a step-like pattern in the spreading behavior. As the slip length b becomes smaller, it approaches a no-slip condition. The results at slip lengths of 10^{-3}m and 10^{-5}m are similar, indicating that the simulation is nearing a no-slip condition at these smaller lengths.

For the cases with slip lengths b of 1m, 0.1m, and 0.001m, sensitivity studies with different particle resolutions of $\Delta x = 3\text{mm}$, 4mm , 5mm are conducted and results are depicted in Fig. 13. The results demonstrate consistent outcomes between the various spatial resolutions, with very similar initial tendencies and overall trends observed.

In Fig. 14, the time evolution of the corium spreading front is

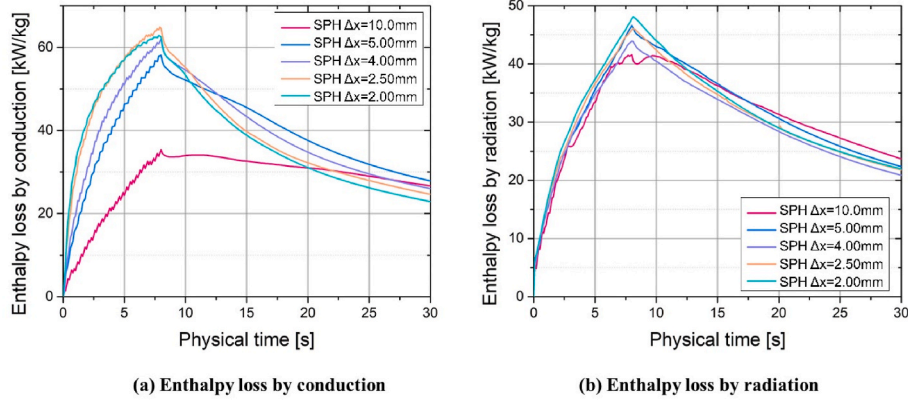


Fig. 11. Vulcano VE-U7: Time evolution of the amount of enthalpy loss due to different heat removal mechanism with different particle resolutions: (a) Conduction (b) radiation.

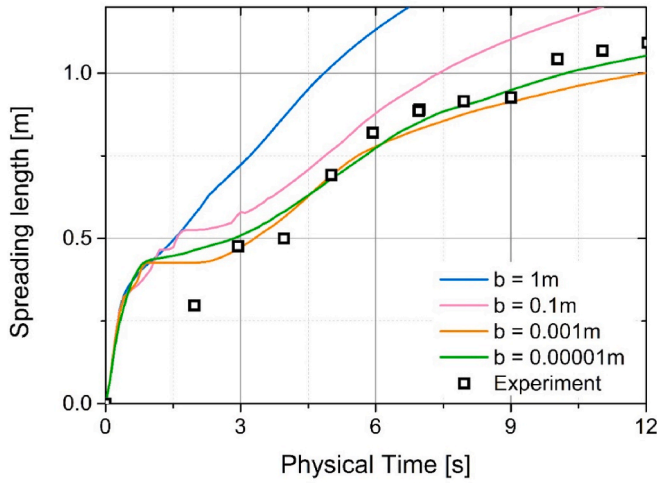


Fig. 12. FARO-L26S: Time evolution of the position of the spreading front of corium with different slip length.

depicted alongside experimental data. Our results are plotted for the case with a slip length of 10^{-5} m in Fig. 13, which is the closest to no-slip among the earlier results. The initial abrupt halt in the spread of corium, caused by rapid solidification, is followed by a resurgence in movement due to continuous mass flow from behind. This dynamic creates a step-like pattern in that is evident in both the experiment and our simulation. However, in this study we utilized simplified geometry information as in previous literature [21], so the initial trend could not be perfectly consistent with experimental results. Instead, analytic solution is

considered as reference as in Fig. 14. Given the high bulk temperature and continuous injection of mass in the experiment, the initial phase is predominantly governed by the bulk viscosity and might be similar with isothermal spreading with constant initial viscosity. The analytic solution for isothermal spreading with constant viscosity can be computed as follows [53]:

$$r_f(t) = \left(\frac{\varphi}{2\pi}\right)^{-\frac{3}{8}} \theta_f \left(\frac{gQ^3}{3\mu_0}\right)^{\frac{1}{8}} (t)^{\left(\frac{1}{8}\right)(3\alpha-\beta+1)}, \quad (37)$$

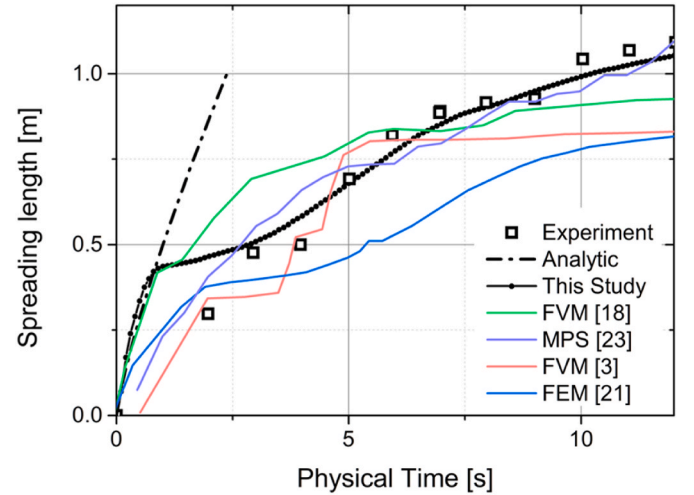


Fig. 14. FARO-L26S: Time evolution of the position of the spreading front of corium compared with results from other literatures.

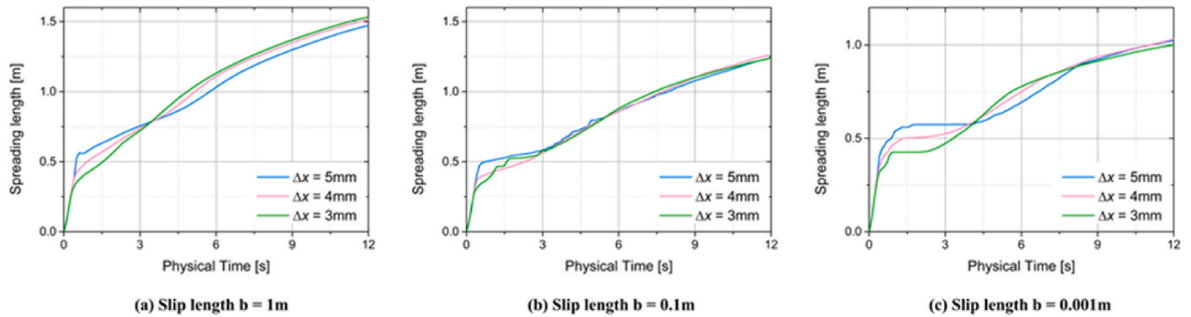


Fig. 13. FARO-L26S: Time evolution of the position of the spreading front of corium with different slip lengths (a) 1m (b) 0.1m (c) 0.001m with different particle resolutions.

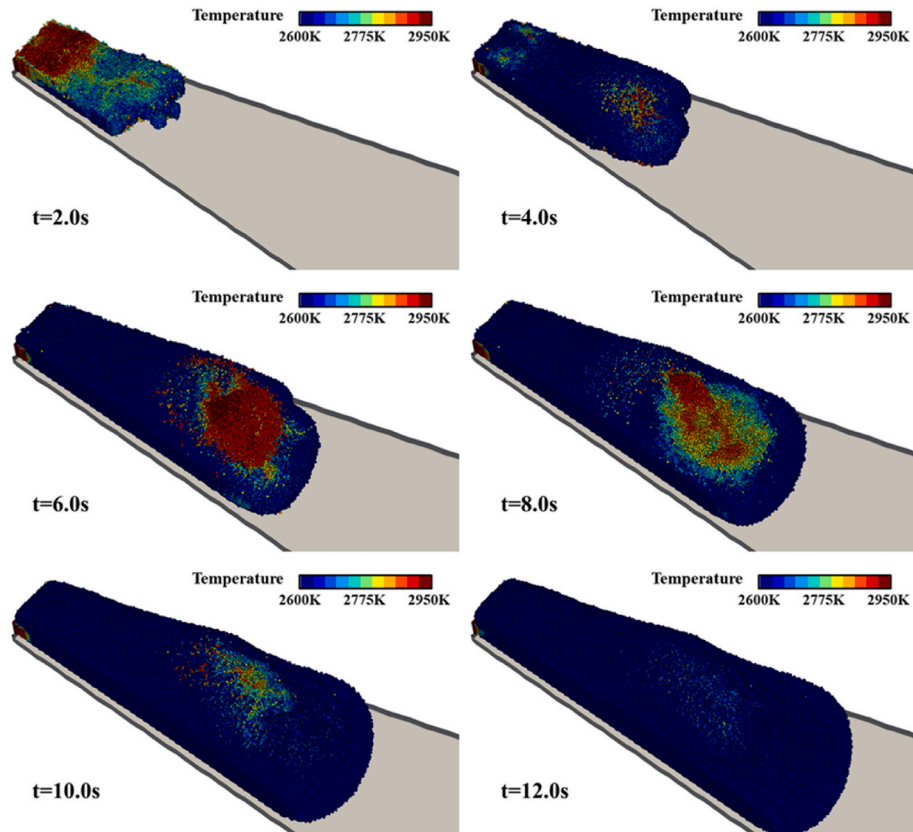


Fig. 15. FARO-L26S: Time Evolution of Corium Progression (The colors of the particles represent their temperatures). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

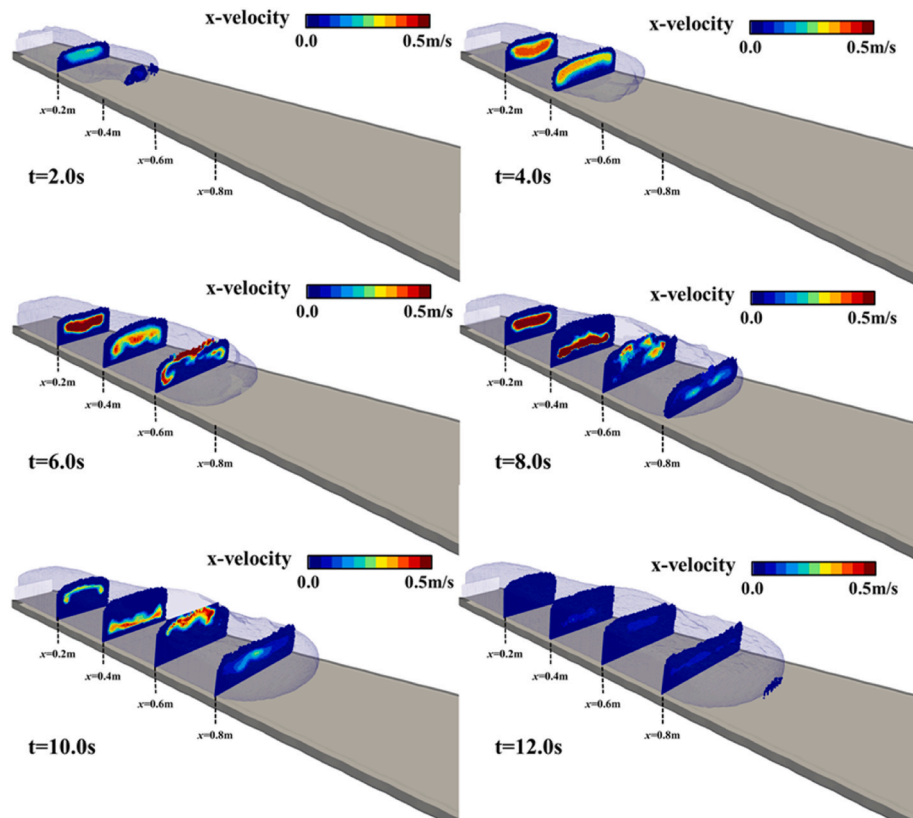


Fig. 16. FARO-L26S: Time Evolution of Velocity Contours at $x = 0.2m$, $0.4m$, $0.6m$, and $0.8m$ Positions at Various Time Instants.

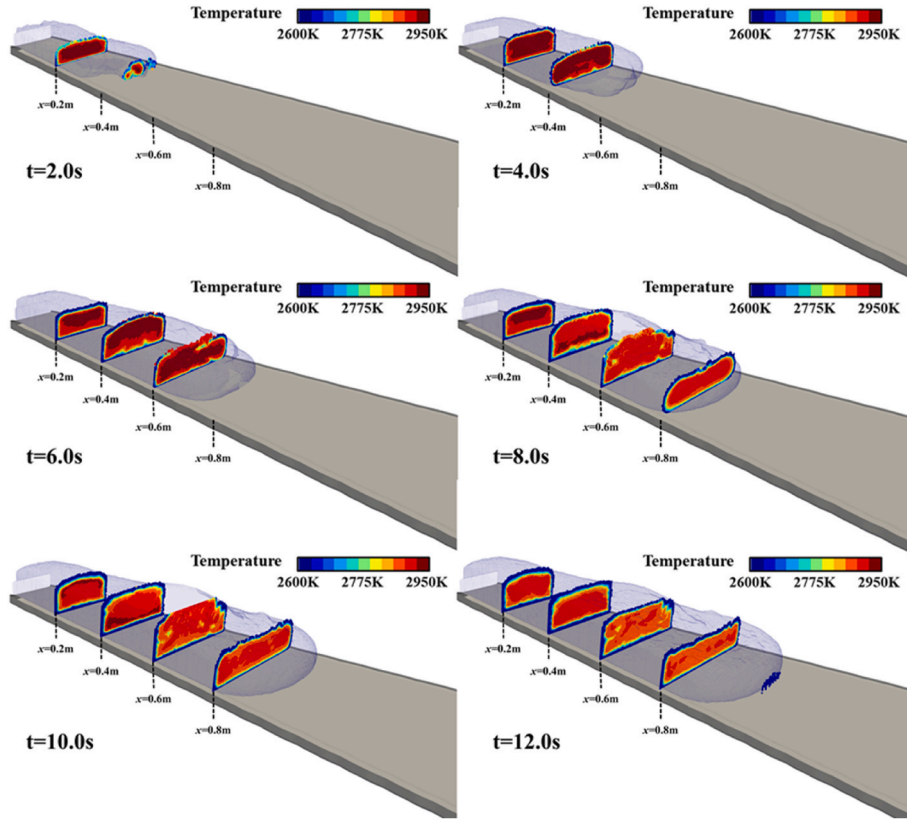


Fig. 17. FARO-L26S: Time Evolution of Velocity Contours at $x = 0.2\text{m}$, 0.4m , 0.6m , and 0.8m Positions at Various Time Instants.

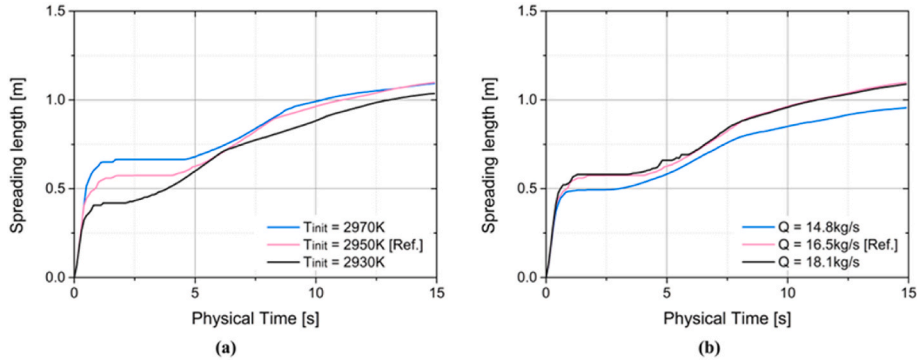


Fig. 18. FARO-L26S: Time Evolution of the Position of the Spreading Front of Corium with Different Inlet Conditions of (a) Temperature and (b) Mass flux.

$$\theta_f := \left[\frac{9}{7} \pi b_1 \left(\frac{1}{2} + \frac{1}{5} b_2 \right) \right]^{-\frac{3}{8}}, b_1 := \left[\frac{3}{8} (3\alpha - \beta + 1) \right]^{1/3}, b_2 := \frac{1}{6} \left(\frac{3\alpha + \beta - 1}{3\alpha - \beta + 1} \right) \quad (38)$$

In our case, we consider time-independent viscosity and constant volumetric flux so, α and β are 1 and 0, respectively. The angle φ is 17° in our case and the volumetric flux Q is about $0.002 \text{ m}^3/\text{s}$.

As shown in Fig. 14, our initial results align very closely with the analytic expectations compared to other studies. As time progresses, it becomes evident that the overall trend aligns well with the experimental results.

Fig. 15 displays the morphology of the corium melt with particle imaging, and Figs. 16 and 17 illustrates the time evolution of temperature and velocity contours. These figures capture the dynamic nature of corium flow in the FARO-L26S experiment, reflecting the complex

interplay between thermal properties and fluid dynamics. A key difference from the VULCANO experiment is the presence of a narrow mushy zone, which allows for rapid phase changes even in the upper region, leading to a quick cessation of movement. As a result, a section where the spreading length no longer progresses is formed. Subsequently, as mass flow continues to be supplied, it pushes the corium ahead, leading to a slow resumption of spreading. This detailed visualization aids in understanding the different behaviors associated with the corium spreading under different mushy zone ranges.

Uncertainties in inlet conditions, specifically temperature and mass flow, were evaluated through a series of sensitivity analyses conducted with a slip length b of 10^{-3} m and a particle spacing of 5 mm as in Fig. 18. The inlet temperature was adjusted to $\pm 20 \text{ K}$ from a reference temperature of 2950 K , and the mass flow rate was varied by $\pm 10 \%$ to observe the effects on the simulation results. Across all cases, the overall progression patterns were similar. Since all the cases were above the melting temperature, the bulk viscosity remained identical, leading to

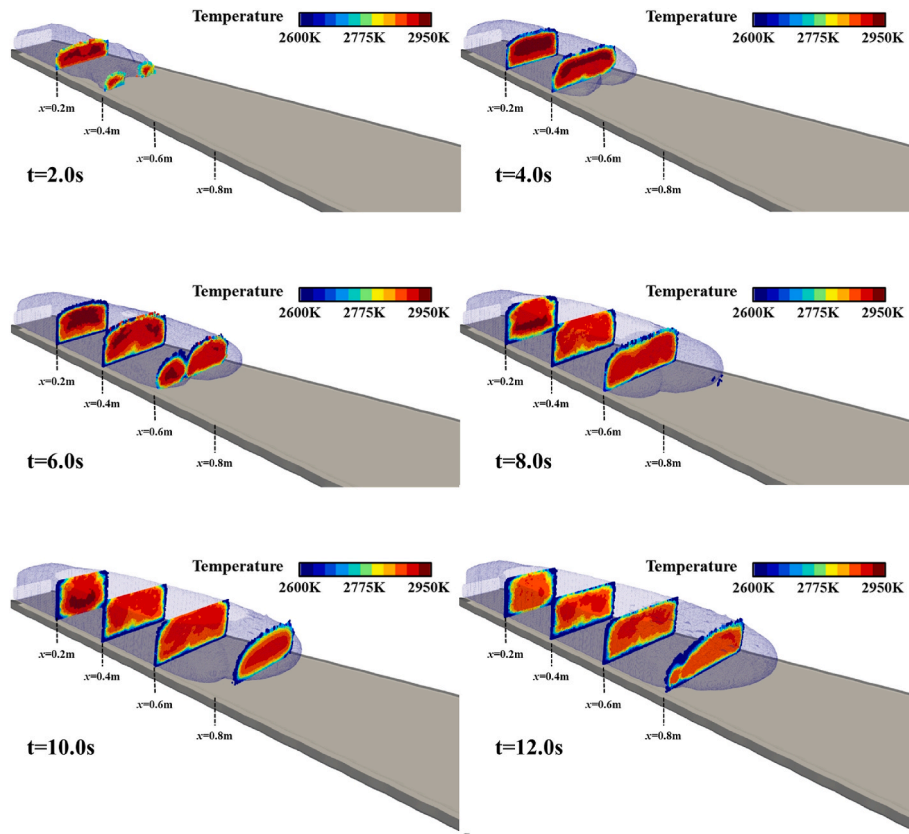


Fig. 19. FARO-L26S: Time Evolution of Temperature Contours at $x = 0.2\text{m}$, 0.4m , 0.6m , and 0.8m Positions at Various Time Instants at Inlet Temperature 2930 K with Particle Resolution of 5 mm .

very similar initial trends. However, there were slight differences in the timing of when the corium became stuck due to cooling. These differences in timing subsequently impacted the final spreading lengths.

To more precisely observe the dynamics of corium progression, the time evolution of temperature contours is scrutinized as in Figs. 19 and 20 for the cases with the lowest and highest inlet temperatures, respectively. As illustrated in Figs. 19 and 20, before 2 s , a significant amount of bulk corium moves swiftly and solidifies at the forefront. In the case where the inlet temperature is set at 2930 K in Fig. 20, additional masses of corium continue to rapidly enter the system due to the high temperature, quickly solidifying and forming a barrier. This barrier then necessitates more time to accumulate enough mass that can push past it. Conversely, when the inlet temperature is set at 2930 K in Fig. 19, the margin to the solidification point is reduced, leading to quicker solidification of the corium and forming a lower and flat barrier. As a result, subsequent masses overlap and flow over the pre-solidified corium with ease, leading to a shorter duration of constant spreading length compared to scenario where the temperature is at 2970 K .

In cases where the mass flow was adjusted, an increase of about 10% in mass flow had minimal impact on the results, while a decrease in mass flow led to a quicker solidification and sticking of the corium. This quicker sticking occurred because the reduced volume of incoming bulk was less effective in pushing forward the stuck corium. However, aside from the significant differences in flow rate, the overall progression trends and dynamics showed no major discrepancies. However, for both sensitivity tests involving variations in inlet temperature and mass flow rate, the variance in the final spreading length was shown to be about 10% with reference scenario.

5. Conclusion

In this study, corium spreading phenomena is deeply investigated

using SPH method. The explicit incompressible SPH (EISPH) approach is utilized to effectively handle fluid dynamics and heat transfer processes. To enhance stability and precision, the governing equations are solved using a quasi-Lagrangian formulation complemented by an upwind scheme. Furthermore, an optimized particle shifting technique is implemented to precisely control the shifting velocity of particles at the free surface and nearby areas. To efficiently manage the highly viscous nature of corium during phase changes, an implicit viscosity solver has been integrated. The entire algorithms are optimized and parallelized on a single GPU architecture to enhance computational efficiency.

Two distinct corium spreading phenomena are examined: the VULCANO VE-U7 experiment, characterized by a wide mushy zone, and the FARO L26S experiment, which has a narrow mushy zone. The differing physico-chemical characteristics of the corium in these experiments result in markedly different spreading behaviors.

In the VULCANO VE-U7 scenario, the extensive mushy zone allows for a gradual increase in viscosity as temperatures rise, facilitating very smooth fluid flow. The numerical analysis of the time-dependent spreading length aligns closely with experimental data. Furthermore, assessments of local temperatures and overall heat removal were conducted to demonstrate both local and global convergence within the study.

Conversely, in the FARO L26S experiment, the restricted mushy zone leads to a swift increase in viscosity as temperatures decrease, producing a step-like pattern in the experimental data. These qualitative characteristics are effectively captured in the numerical investigation, and our results successfully capture the rapid progression at the initial stage, aligning well with the analytic solution. The temporal evolution of shape and temperature contours in this scenario reveals more complex patterns than those observed in the VE-U7 experiment, emphasizing the unique dynamics of the flow resulting from the differing characteristics of the mushy zones. Both quantitative and qualitative results in our

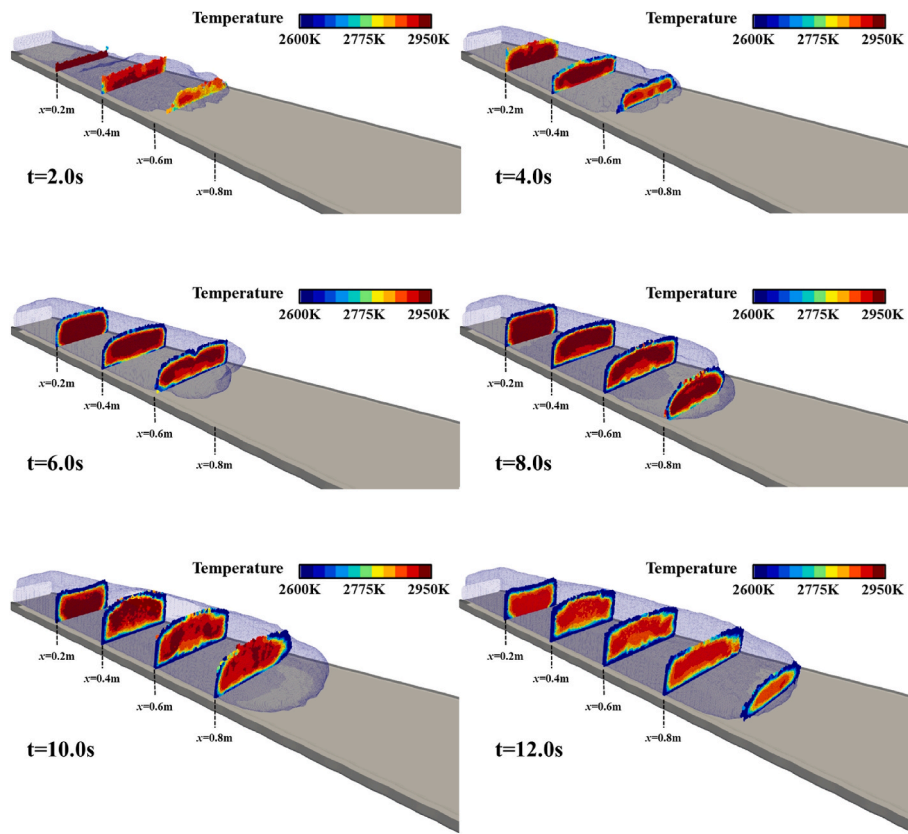


Fig. 20. FARO-L26S: Time Evolution of Temperature Contours at $x = 0.2\text{m}$, 0.4m , 0.6m , and 0.8m Positions at Various Time Instants at Inlet Temperature 2970 K with Particle Resolution of 5 mm .

study underscore the capability and potential of our method to accurately model and analyze these complex corium spreading phenomena.

CRedit authorship contribution statement

Hee Sang Yoo: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Conceptualization. **Eung Soo Kim:** Writing – review & editing, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Eung Soo Kim reports financial support was provided by National Research Foundation of Korea and Nuclear Safety and Security Commission.

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